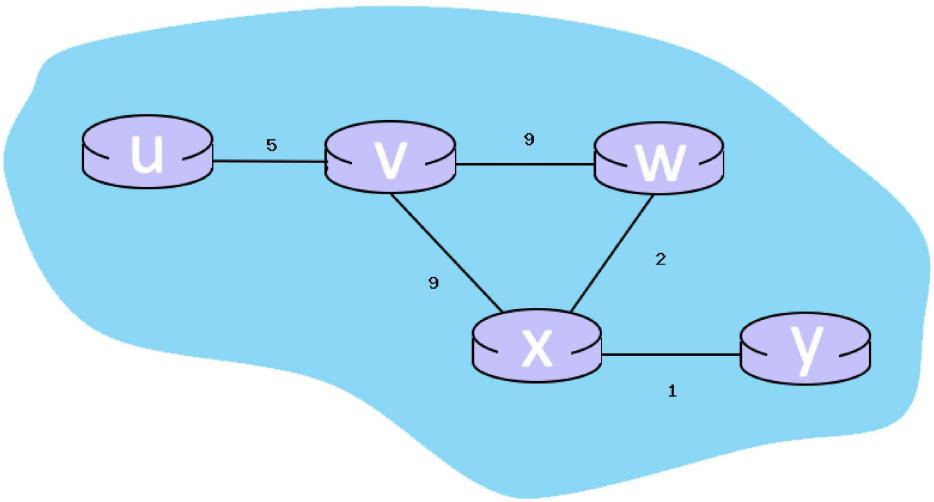
INTERACTIVE END-OF-CHAPTER EXERCISES

BELLMAN FORD DISTANCE VECTOR ALGORITHM (FOR COMPUTING LEAST COST PATHS)

Consider the 6-node network shown below, with the given link costs:



QUESTION LIST

1. When the algorithm converges, what are the distance vectors from router 'X' to all routers? Write your answer as u,v,w,x,y

2. What are the initial distance vectors for router 'V'? Write your answer as u,v,w,x,y and if a distance is ∞, write 'x'

3. The phrase 'Good news travels fast' is very applicable to distance vector routing when link costs decrease; what is the name of the problem that can occur when link costs increase?

SOLUTION

1. When the algorithm converges, router X has distance vectors (u,v,w,x,y) = (14,9,2,0,1)

2. The initial distance vectors of router V are: (u,v,w,x,y) = (5,0,9,9,x) where x is ∞

3. It is called the 'Count to Infinity' problem.

QUESTION 1 OF 3

When the algorithm converges, what are the distance vectors from router 'X' to all routers? Write your answer as u,v,w,x,y

Answer

CHECK HINT SKIP

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